



GE Baseline Security Center Cybersecurity System Test Procedure Core Security Module (CSM)

PRODUCT	BASELINE SECURITY CENTER (BSC)	REVISION	1.1
SITE	Customer Name – Site Name	DATE	MM/DD/YYYY
		PAGES	25





Table of Contents

1.	INTRODUCTION	6
2.	CYBERSECURITY TEST PLAN	7
2.1	SYSTEM READINESS & NETWORK CONNECTIVITY VERIFICATION	7
2.1.1	Ensure System Readiness	7
2.1.2	Verify Control Server Connectivity to the Controls Network Using Echo Request / Ping	7
2.1.3	SMB Connectivity	8
2.1.4	RDP Access	8
2.2	ACCOUNT MANAGEMENT	9
2.2.1	Default Windows Accounts	9
2.3	SYSTEM BACKUP & RECOVERY	10
2.3.1	Backup Configuration	10
2.3.2	System Backup Task Execution	10
2.3.3	Backup Repository Capacity	11
2.4	LOG AGGREGATION.....	12
2.4.1	Windows Event Logs	12
2.4.2	Network Device Logging.....	12
2.5	NETWORK DEVICE MANAGEMENT	13
2.5.1	Secure Access to Network Devices.....	13
2.6	APPLICATION CONTROL (McAfee EPO ORCHESTRATOR)	16
2.6.1	Secure Access to McAfee ePO Orchestrator Management Server	16
2.6.2	System Configuration – Asset Organization	16
2.6.3	Blocking Execution of Unapproved Programs & Files	16
2.6.4	Continued Application Control Enforcement Even After Loss of Central Management Server.....	17
2.6.5	System Configuration – Application Lockdown Status.....	17
2.6.6	System Configuration – Trusted Updater.....	17
2.7	SIEM.....	18
2.7.1	SIEM Dashboards – GE Default Dashboard & NERC-CIP Dashboards	18
2.7.2	SIEM Device Configuration – Verify Asset Listing.....	18
2.7.3	SIEM Device Configuration – Verify Data Sources	19
2.7.4	SIEM Device Configuration – Verify Event Logging	19
2.7.5	SIEM Report Generation	19
2.8	CENTRALIZED PATCH MANAGEMENT & DEPLOYMENT	20
2.8.1	System Configuration – Asset Organization	20
2.8.2	Patch Status – Control Server & Physical Systems	20
2.8.3	Patch Management Process – Customer Documentation Hand-Off	21
2.9	GENERAL SYSTEM FUNCTIONALITY	22
2.9.1	Time Synchronization	22
2.9.2	Customer Documentation Hand-Off	23



3.	APPROVALS / SIGN-OFF	24
3.1	ISSUE REPORT WORK SHEET.....	24



Revision History

Revision	Date	Editor	Revision Summary
1	05/14/2019	Jonathan Couch	Initial document release
1.1	08/02/2021	Michael Berube	Updates to test case content



1. Introduction

The purpose of this Cybersecurity System Test (CST) is to verify the proper operation of all equipment for the **Site Name-Customer Name** project as a system prior to shipment. The CST procedure verifies:

- Proper system integration of the GE Baseline Security Center (BSC) security appliance
- Expected performance of the GE BSC when integrated with the GE Control Server, GE traditional thick-client HMI equipment, available network equipment and other available GE provided equipment

Procedure Deliverables:

1. Signed, completed CST plan
2. Supporting documentation highlighting any outstanding / open items at the conclusion of the CST
3. Network port scan results of accessible network segments
4. GE BSC system configuration documentation (standard ports, services, service accounts and system credential information)



2. Cybersecurity Test Plan

The following tests constitute the cybersecurity test plan associated with the implementation of GE's Baseline Security Center in Control Server system architecture environments.

2.1 System Readiness & Network Connectivity Verification

The following test items are designed to verify general system readiness, availability, and connectivity.

2.1.1 Ensure System Readiness

Description:	Verify all physical hosts, virtual machines, network devices, and any available infrastructure equipment (ATS switch, etc.) are available for the test, are connected and are powered on. (Refer to site's 4108 network topology diagram)			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Perform review of 4108 drawing			

2.1.2 Verify Control Server Connectivity to the Controls Network Using Echo Request / Ping

Description:	Verify that the GE Control Server (hypervisor and all VMs) components are properly connected to the controls network by pinging a sample of the following, as applicable:			
	Item Description		See Asset Checklist for reference	
	Control Server physical hosts			
	Control Server VMs			
	Additional physical HMIs/Historians			
	Available network devices			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	See Asset Checklist for reference -			



2.1.3 SMB Connectivity

Description:	Verify that the network file access to the default administrative share (\\<SYSTEM_NAME>\C\$) can be established over the following connections: 1. MC3 (Host for BSC Apps) -> Control Server VMs and any additional physical systems running a Windows OS environment. 2. Control Server VMs / physical systems -> MC3			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	See Asset Checklist for reference -			

2.1.4 RDP Access

Description:	Verify and document hosts that allow RDP access. Disable RDP access where applicable if requested by the customer. List of Hosts:			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	See Asset Checklist for reference –			



2.2 Account Management

The following test items are designed to verify functionality and configuration of the account management controls in place on the system.

2.2.1 Default Windows Accounts

Description:	Verify that the following default Windows user accounts have been deleted or disabled from:															
	• each of the listed domains • a selected local HMI															
	<table border="1"><thead><tr><th>User Account Name</th><th>Local HMI</th><th>HMI Domain</th></tr></thead><tbody><tr><td>Administrator</td><td></td><td></td></tr><tr><td>DeafulyAccount</td><td></td><td></td></tr><tr><td>Guest</td><td></td><td></td></tr></tbody></table>				User Account Name	Local HMI	HMI Domain	Administrator			DeafulyAccount			Guest		
	User Account Name	Local HMI	HMI Domain													
	Administrator															
DeafulyAccount																
Guest																
Local HMI Tested: _____																
Sign-Off:	Test Engineer:	Michael Berube	Date:													
	Customer Representative:		Date:													
Comments:																



2.3 System Backup & Recovery

The following test items are designed to verify functionality and configuration of the centrally managed backup and restore capabilities provided by the GE BSC 2.0 security system.

2.3.1 Backup Configuration

Description:	Verify the following: - Automated backups are configured for each Windows and Linux OS environment to be managed by the GE BSC 2.0 system. - Backup type / contents and backup schedule with the customer representative. - Ensure backup retention period meets customer requirements if customer requirement deviates from standard BSC retention periods.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	See Asset Checklist for reference -			

2.3.2 System Backup Task Execution

Description:	Verify a system backup exists for each type of system asset supplied by GE: 			
	Asset Type		See Asset Checklist for reference	
	Control Server VMs			
	Additional physical HMIs/Historians			
	BSC VMs			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	See Asset Checklist for reference -			



2.3.3 Backup Repository Capacity

Description:	Verify that the backup repository is provided with sufficient capacity to store, at minimum: • Two (2) generations of backups of all the managed assets, or • The number of backups / retention period specified by any existing customer requirements designed to supersede the BSC default backup retention capacity			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.4 Log Aggregation

The following test items are designed to verify functionality and configuration of the log aggregation capabilities provided by the GE BSC security appliance.

2.4.1 Windows Event Logs

Description:	Verify the McAfee Log Manager is properly configured to collect Windows Event Logs from each managed asset.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.4.2 Network Device Logging

Description:	Verify that installed network devices capable of logging via Syslog are configured to forward log events to the McAfee SIEM.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Refer to Asset Checklist – Network Devices Tab -			



2.5 Network Device Management

The following test items are designed to verify functionality and configuration of the Network Device Management capabilities provided by the GE BSC security appliance.

2.5.1 Secure Access to Network Devices

Description:	Using the PuTTY utility to connect to selected switches, perform the following verification:			
	• Verify that connections to network devices to obtain network baseline / configuration data is established via SSH. • Demonstrate the SSH whitelist in the switch configuration • Demonstrate that telnet connections are blocked Network Device IP(s) Tested: _____			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.5.2 RADIUS Functionality.

Step	Description	Verified	Comments and Observations
1	Review RADIUS in Active Directory on DC1 and DC2 to verify all switches are entered.		
	Date: _____	Initials_____	
2	Verify switches can be accessed with the Active Directory account 'GESwitch'.		
	Date: _____	Initials_____	
3	Disable RADIUS service on DC1 and DC2 and attempt to log into a switch. When failure occurs, verify switch can be accessed with the local admin account.		
	Date: _____	Initials_____	



2.5.3 Cat Tools

Step	Description	Verified	Comments and Observations
1	Verify all network switches are listed in CatTools.		
	Date: _____	Initials: _____	
2	Verify all switches can be backed up in CatTools		
	Date: _____	Initials: _____	
3	Use the reports function to verify any switch changes can be documented.		
	Date: _____	Initials: _____	
5.2.4	Verify CatTools uses the GESwitch account for accessing the Cisco switches.		
	Date: _____	Initials: _____	
5.2.5	Verify passwords in CatTools are not stored in plain text		
	Date: _____	Initials: _____	





2.5.4 Switch Logging to SIEM

Step	Description	Verified	Comments and Observations
1	Verify successful and unsuccessful login attempts are captured in the SIEM.		
	Date: _____	Initials: _____	
2	Verify switches store logs locally.		
	Date: _____	Initials: _____	
3	Verify switch time is set for proper NTP source		
	Date: _____	Initials: _____	



2.6 Application Control (McAfee ePO Orchestrator)

The following test items are designed to verify functionality and configuration of the centrally managed application control capabilities provided by the GE SecurityST security appliance.

2.6.1 Secure Access to McAfee ePO Orchestrator Management Server

Description:	Verify that access (via direct URL entry and the browser bookmark) to the McAfee ePO Orchestrator Management Server URL is via HTTPS not HTTP. McAfee ePO can be accessed via direct URL entry at https://mc3:8448/core/orionSplashScreen.do			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.6.2 System Configuration – Asset Organization

Description:	Verify that all managed assets/systems are listed in the McAfee ePO Orchestrator Management Server console and are appropriately grouped / organized.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.6.3 Blocking Execution of Unapproved Programs & Files

Description:	Verify that the McAfee ePO Endpoint client will not open or execute a file that is not on the approved list. Attempting to execute a new unapproved application will have the following results: • Application will fail to run and will generate an error message • An entry will be added to the “Execution Denied” report within EPO Orchestrator 			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.6.4 Continued Application Control Enforcement Even After Loss of Central Management Server

Description:	Verify that the deployed application control policy continues to be enforced on managed assets even after the McAfee ePO Orchestrator Management Server is unavailable. To simulate – On the MC3 server, stop the “McAfee ePolicy Orchestrator 5.10.0 Application Server & McAfee\Policy Orchestrator” service from the Service management console. And then verify the functionality.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.6.5 System Configuration – Application Lockdown Status

Description:	Verify that all managed assets/systems are configured with Solid Core in Observe Mode prior to shipment to site for commissioning. This will avoid complications during site installation.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Refer to Asset Checklist			

2.6.6 System Configuration – Trusted Updater

Description:	Verify that the WorkstationST.exe executable is defined as a Trusted Updater and that this setting has been propagated to all the HMI's running ControlST. System Tree – Assigned Policies – Product-Solidcore 8.3.2:App Control, Application Control Rules (Windows) – GE Default Select ControlST under My Rules Left Pane This configuration change ensures that the Static Starter and Exciter screens fully operate.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.7 SIEM

The following test items are designed to verify functionality and configuration of the Security Incident and Event Monitoring (SIEM) capabilities provided by the GE BSC security appliance.

2.7.1 SIEM Dashboards – GE Default Dashboard & NERC-CIP Dashboards

Description:	1) Demonstrate the following widgets are displayed on the GE Default dashboard & are populated with event data: • Windows Device Events • Network Device Events • VMWare Device Events • Event Summary • Event count by Host • Events 2) Open the NERC-CIP 002 Dashboard and confirm the widgets are populating with event data			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.7.2 SIEM Device Configuration – Verify Asset Listing

Description:	Verify that all applicable assets are listed in the McAfee SIEM appliance, and are populated within their respective Asset Groups defined below: • VMWare • Windows • Network Devices			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.7.3 SIEM Device Configuration – Verify Data Sources

Description:	Verify that all applicable data sources are listed in the McAfee SIEM appliance, and the correct properties are assigned to each one.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.7.4 SIEM Device Configuration – Verify Event Logging

Description:	Verify that an event is displayed in the Windows Login Events widget of the GE Default Dashboard after a successful or invalid login attempt.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.7.5 SIEM Report Generation

Description:	Verify that a selected report can be generated and contains relevant event data.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.8 Centralized Patch Management & Deployment

The following test items are designed to verify functionality and configuration of the centrally managed patch deployment capabilities provided by the GE BSC security appliance.

Note: This section is only applicable in situations where the centralized patch management and deployment solution is included with the BSC security appliance.

2.8.1 System Configuration – Asset Organization

Description:	Verify that all managed assets/systems are listed in the Ivanti Patch for Windows management console and are appropriately grouped / organized.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Refer to Asset Checklist -			

2.8.2 Patch Status – Control Server & Physical Systems

Description:	Verify the latest validated patches for installation on Windows OS environments have been applied to the Control Server Windows-based virtual machines and any Windows-based physical HMIs/Historians. Windows assets • Open Ivanti Patch for Windows management console • Run a scan of desired Asset Group Date of Patch Disc Applied: _____ (Must login as Domain Admin)			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Refer to Asset Checklist -			



2.8.3 Patch Management Process – Customer Documentation Hand-Off

Description:	Provide hand-off materials to the customer representative documenting GE BSC standard procedure for obtaining and applying validated patches as part of the Power Patch Validation Program. • DisPatch Web Portal User's Guide • BSC Windows Patching Guide 			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				



2.9 General System Functionality

The following test items are designed to verify general system functionality and configuration of the GE BSC security appliance not captured elsewhere.

2.9.1 Time Synchronization

<



	ENMELM			
	EDGE1SVR			
	HC1*			
	HW1*			
	* Not all VMs may be present based on the type of control solution being provided as part of the project. For example, BU1, BU2 and BU3 will only be present in environments where a GE Control Server is being provided, and the customer has opted to include the Acronis backup and recovery solution.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:				

2.9.2 Customer Documentation Hand-Off

Description:	Provide hand-off materials to the customer representative documenting GE BSC standard ports, services, service accounts and system credential information, as applicable.			
Sign-Off:	Test Engineer:	Michael Berube	Date:	
	Customer Representative:		Date:	
Comments:	Upload all relevant docs to GE Box folder for this project.			



3. Approvals / Sign-Off

_____ GE BSC cybersecurity factory system test completed and approved with no outstanding / open items

_____ GE BSC cybersecurity factory system test completed and approved with outstanding / open items still to complete (refer to documented punch list / action items list below)

3.1 Issue Report Work Sheet

Issue #	Issue Description	Report Date	Assigned To	Expected Closure Date	Status
1					
2					
3					
4					
5					
6					
7					
8					



Issue #	Issue Description	Report Date	Assigned To	Expected Closure Date	Status
9					
10					
11					
12					
13					
14					
15					

Signed for:
GE Cyber Security Representative

Signature: Michael Berube

Name (Printed): Michael Berube

Date: _____

Signed for:
Customer Representative

Signature: _____

Name (Printed): _____

Date: _____